

Functional electrical stimulation-based cycling assisted by flywheel and electrical clutch mechanism: A feasibility simulation study



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HIGHLIGHTS

- A flywheel and electrical clutch assist mechanism is used in FES-cycling exercise.
- Fuzzy logic is used to control stimulation intensity on quadriceps in FES-cycling.
- The new mechanism suppresses speed fluctuation and reduces the cadence error.
- The new assist mechanism promotes prolonged FES-cycling and extended work rate.
- The fatigue monitor shows 14%–17% delay in muscle fatigue using the new mechanism.

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ABSTRACT

A new assist mechanism, represented by a flywheel and an electrical clutch, is developed and evaluated, in simulation studies, to assist paralysed legs during functional electrical stimulation (FES)-based cycling exercise in a closed-loop control configuration. The flywheel is engaged and disengaged, by the clutch, to assist or retard the cycling when necessary. The flywheel engages with the crank to absorb the surplus energy, produced by stimulating the leg, store it as kinetic energy and slow down the movement. Also, it engages again to use the same stored energy to assist the leg and speed up the cycling. A comparative assessment of FES-cycling, using fuzzy logic control, is carried out with and without the new assist mechanism. Clinically recorded data is used to derive a force-drop indicator for assessment purposes. Although the stimulation intensity is slightly increased, the indicator showed 14%–17% muscle fatigue delay with the new mechanism as compared with cycling without assistance. This mechanism is promoting prolonged FES-cycling and increased work rate for paraplegics by delaying the occurrence of muscle fatigue.

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1. Introduction

Functional electrical stimulation (FES) is a technique of supplying a train of electrical stimuli to trigger nerves of paralysed muscles to provide muscle contraction and produce useful functional movement [1]. Several FES-based devices have been developed and utilized for therapeutic and function restoration purposes. For example, the pacemaker, as a heart pulse regulator, is used by more than 500,000 people every year [2]. Also, FES has been utilized to provide lower extremities' movement for people with complete and incomplete spinal cord injuries in an attempt to restore locomotion through different exercises such as walking, standing and cycling. Muscle stimulation with FES can be done either by using implanted electrodes [3,4] or by surface electrodes mounted on the

skin. Stimulation with surface electrodes is easier, cheaper, non-invasive, and has no potential of infection as compared with other types [5].

FES-based cycling is a type of exercise that employs FES signals to stimulate leg muscles of paralysed people in a specific sequence to perform pedalling motion. FES-cycling is more beneficial for a disabled person than weight lifting, although it provides smaller increase in muscle size. The cardiac output of paralysed individuals during weight lifting induced by FES produces 7 l/min while the cardiac output has been shown to rise to 15 l/min during FES-cycling exercise [6–8]. Also, several studies have shown that continuous FES-cycling exercise for paralysed people increases the cardiovascular and cardiorespiratory fitness, blood circulation in lower limbs, reverse muscle atrophy, prevention of bone loss as well as it improves the self-image of the disabled [9–13].

One of the handicaps that affect the performance and the smoothness of FES-assisted cycling is the dead points, i.e. the two points on the pedalling cycle at which it is difficult to produce sufficient torque to rotate the crank. By means of inertia and a complex

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interplay of muscle actions, healthy people can overcome these points, while it is difficult to produce such muscle actions by means of FES because the muscles involved are too deep to be stimulated by surface electrodes [14].

Previous studies have shown several attempts to overcome the dead points and produce smooth, i.e. with no abrupt change in crank velocity, FES-cycling for paraplegics. These have included improving the ergometer's mechanical design using different assist mechanisms. Some researchers have utilized an auxiliary motor to help the leg pass the dead spots and continue the cycling in case muscle fatigue occurs, and also to retard the movement by imposing braking when necessary [15–17]. Other researchers have modified the five-bar linkage of a tricycle into a four-bar linkage by using a coupler and a lever arm to eliminate the dead spots during cycling [14,18] with a drawback of eliminating the freewheeling of the tri-cycle. While Ref. [6] recruited a non-paralysed healthy person, using a modified tricycle of two side-by-side seats, to provide pedalling assistance for a paraplegic to propel the vehicle, pass the cycling dead spots and continue the cycling in case of muscle fatigue, others have added an arm-crank to the ergometer to assist the leg and provide a hybrid exercise mechanism [5].

Besides the assist mechanisms, researchers have stimulated different combinations of muscles to provide leg extension and flexion action. Some researchers have stimulated the quadriceps and hamstring of both legs [19]. Others have used the quadriceps, hamstring and the gluteus maximus muscles [5,6,17,20,21]. While others have used the quadriceps, hamstring and the gastrocnemius muscles [15] during FES-cycling exercise.

Controlling the movement of paralysed limbs with FES using open-loop control strategy is particularly difficult. As several parameters differ from person to person, such as muscle response to FES, skin sensitivity and muscle's training condition, the stimulation parameters applied in open-loop systems are specific for a single user and may not produce the same performance with other persons [22,23]. Also, the procedures of determining the stimulation parameters are time consuming; trials lasted for 20–45 min to find the optimal parameters for the leg, by stimulating quadriceps only, to follow a desired trajectory [24]. Moreover, the open-loop approach cannot account for unforeseen conditions such as muscle spasm and mechanical disturbances. For these reasons researchers have focused on utilizing closed-loop control strategies to overcome the aforementioned problems [19,20,25–29]. To reduce the possible mechanical problems that might occur during FES-cycling due to several electrode wirings and in an attempt to provide more comfortable exercise by reducing the pre-cycling preparations required for locating the electrodes at their optimal locations over the skin to get optimal muscle response, Ref. [30] produced stimulation patterns, to perform coordinated FES-assisted cycling movement, based on stimulating single muscle group, the quadriceps, of each leg.

The flywheel, as an energy storage device, has been widely used in many commercial FES-cycling ergometers. It has been used to provide smoothness to the cycling and help pass the cycling dead spots for individuals able to pedal under loads [21]. Usually, disabled people encounter difficulties to pedal the crank of the ergometer due to weak leg muscles [31]. Although the flywheel is effective in assisting the cycling and reducing the energy expenditure [32], from other point of view, a fix-geared flywheel imposes extra load on the crank which in turn makes it harder for individuals of weak muscles to generate sufficient force and overcome the inertia to drive the flywheel without external assistance [21].

A hybrid kinetic energy recovery mechanism consisting of a flywheel, electrical clutch and continuously variable transmission (CVT) was designed for formula one motorsports in 2009 for the purpose of fuel consumption [33]. The flywheel is used to store the kinetic energy in the vehicle during braking, and later reuse

the same stored energy to accelerate the vehicle. The test results showed the ability of the system to save up to 21% of the driving energy of the vehicle.

The use of such a mechanism has not been reported with FES-cycling application. In this work, it is aimed to implement a similar mechanism, represented by a flywheel and an electrical clutch, in an attempt to save the excessive energy in the system and make use of it to prolong the exercise. The flywheel as an energy storage device is to be used to absorb and store the surplus of kinetic energy in the system in order to retard the movement when necessary, and later reuse the stored kinetic energy to help the leg pedal and speed up the movement. The use of a clutch is necessary to provide the flywheel's engagement and disengagement to/from the crank (shaft) of the cycling ergometer.

In this work, for assessment purposes, a comparison between two approaches will be presented; the first scenario is a closed-loop control strategy for FES-cycling by stimulating single muscle group, the quadriceps, without any assist mechanism. The second scenario is a closed-loop control strategy for FES-cycling by stimulating the quadriceps with the aid of the flywheel and electrical clutch mechanism. This work is a developed version of the authors' initial work [34] through the use of fuzzy logic controllers, instead of PID, and an improved, more accurate and easy-to-implement, flywheel engagement mechanism. The new proposed approach, with a similar tracking performance, reduces the overall stimulation intensity on both legs by approximately 30% (from 685 μ s to 475 μ s) which consequently delays muscle fatigue and prolongs the cycling exercise.

2. System description

The use of an accurate model is important from the viewpoint of system simulation to obtain results close to reality as well as from the perspective of control design. In addition to the accuracy of the data used to build the model, the technique or the software utilized to simulate the plant behaviour plays a significant role in obtaining accurate results. In this work a humanoid–bicycle model, equipped with a flywheel and an electrical clutch mechanism, is developed using Visual Nastran 4D (Vn4D) software. The Vn4D software is selected for its ability to combine computer-aided design (CAD), motion and finite element analysis (FEA) in a single modelling system. Moreover, it is equipped with sensors, meters and controllable constraints useful for studying the behaviour of control systems in a simulation platform. Furthermore, the Vn4D software can be easily connected with Simulink/matlab software, hence allowing design and evaluation of control strategies.

2.1. The humanoid model

The quality of the humanoid model depends on the accuracy of the data used to build the model. In this work, the humanoid model is developed using the standard anthropometric human dimensions introduced by Ref. [35] as shown in Fig. 1. The length and the mass of each body segment are expressed as fraction of the overall body height and weight respectively.

The humanoid developed in this work is based on a human body of 1.80 m in height (H) and 70 kg in weight (M). The length and mass of each segment of the developed humanoid model are shown in Tables 1 and 2 respectively.

The centre of mass and the density of each segment were obtained from the same anthropometric data. The centre of mass was essential to determine the shape of each segment, while the density of each segment was used to obtain the volume and consequently the segment's width. Table 3 shows the location of centre of mass, density and volume of each segment of the developed humanoid model.

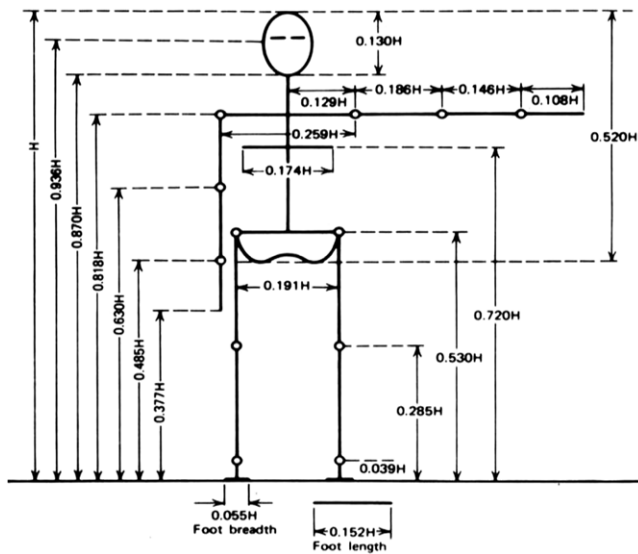


Fig. 1. Standard anthropometric humanoid dimensions [35].

Table 1
Body segment length of the developed humanoid model.

Segment	Segment length, fraction of height (H)	Humanoid model length (m)
Head	0.130 H	0.234
Neck	0.052 H	0.0936
Trunk	0.288 H	0.5184
Upper arm	0.186 H	0.3348
Lower arm (Forearm)	0.146 H	0.2628
Hand	0.108 H	0.1944
Pelvis and Thigh	0.245 H	0.441
Shank	0.246 H	0.4428
Foot height	0.039 H	0.0702
Foot breadth	0.055 H	0.099
Foot length	0.152 H	0.2736

Table 2
Body segment weight of the developed humanoid model.

Segment	Segment mass, fraction of mass (M)	Humanoid model mass (kg)
Head and Neck	0.081 M	5.67
Trunk	0.497 M	34.79
Upper arm	0.028 M	1.96
Lower arm (Forearm)	0.016 M	1.12
Hand	0.006 M	0.42
Pelvis and Thigh	0.100 M	7
Shank	0.0465 M	3.255
Foot	0.0145 M	1.015

Table 3
Body segment's centre of mass, density and volume of the humanoid model.

Segment	Location of centre of mass/Segment length (Proximal)	Density (kg/l)	Volume (m ³)
Head and Neck	1.000	1.11	0.0051
Trunk	0.5	1.03	0.0337
Upper arm	0.436	1.07	0.0018
Lower arm (Forearm)	0.430	1.13	0.0009
Hand	0.506	1.16	0.0003
Pelvis and Thigh	0.433	1.05	0.0066
Shank	0.433	1.09	0.0029
Foot	0.500	1.10	0.0009

Body segments are connected to each other by joints provided in Vn4D software as constraints. The head and neck joints were considered as rigid joints as they have no significant effect on

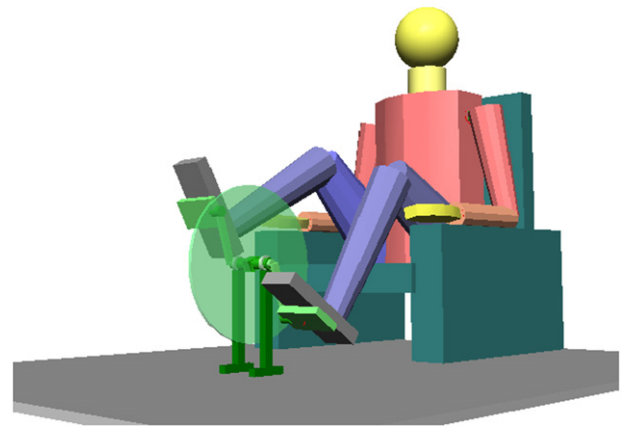


Fig. 2. The developed humanoid-bicycle model with flywheel and clutch mechanism.

the performance of FES-cycling training. Also, the ankle joint that connects the foot with the shank is considered as a rigid joint to represent the ankle-foot orthoses (AFO) used in FES-cycling for safety purposes and to allow full transmission of leg's torque into the crank of the bicycle. The shoulder, elbow, wrist and hip joints are represented by freely revolute joints, while the knee joints of right and left legs are represented by revolute motors in order to be controlled by the torque generated by the quadriceps muscle group of each leg. It is worth mentioning that in this phase of the project, the dimensions of the upper body do not affect the simulation results. However, in future, the humanoid model will be used in studying the effect of hip joint angle on the cycling performance as well as in mobile ergometer scenarios.

2.2. The bicycle model

The bicycle model was also developed using Vn4D software. The dimensions of the bicycle were considered as (pedal: $0.13 \times 0.08 \times 0.02$ m; crank arm: $0.01 \times 0.14 \times 0.02$ m; crank (shaft): 0.01×0.15 m). The specifications of the designed bicycle model were obtained from a real cycling ergometer available in the laboratory at the department of Automatic Control and Systems Engineering, The University of Sheffield. To simulate a more realistic system and obtain more reasonable results, a standard ball bearings friction with rotational coefficient (0.0015) and effective radius (0.01 m) was added to the model.

As an assisting mechanism, a flywheel and an electrical clutch are added to the bicycle model using the same software. The flywheel dimensions used are (radius: 0.2 m, height: 0.01 m, weight: 3.5 kg). To simulate the behaviour of an electrical clutch that is responsible for engaging/disengaging the flywheel with/from the crank (shaft), a rigid constraint, i.e. joint, with on/off operating condition, between the flywheel and the crank (shaft) is implemented in Vn4D software. The engagement and disengagement of the flywheel is to be controlled through an on/off control input via Simulink/Matlab software. The developed humanoid-bicycle model with the flywheel and electrical clutch mechanism is shown in Fig. 2.

2.3. The muscle model

The human musculoskeletal muscles, that are responsible for producing voluntary movement, have been widely described in the literature [36–41]. One of the well-known and frequently used muscle models is that developed by Hill [36]. Since the development of Hill's muscle model several attempts have been made to increase the accuracy of the model by adding further information,

such as adding the tendon effect [38,39] or interpreting the physiological behaviour of the muscle into three parts; muscle activation, which includes calcium dynamic and muscle fatigue phenomena; muscle contraction and segmental dynamics [42,43]. Although muscle models based on the physiological behaviour of the muscle are assumed to be more accurate than others, several parameters are required to be optimized to obtain an acceptable response, which increases the implementation complexity. For this reason, in this work, it is preferred to use the model featured in Ref. [40], which is simple to implement and accurate enough as it has been derived from data obtained experimentally from paraplegics and healthy subjects. The model is represented by a single-pole transfer function that mimics the behaviour of the quadriceps muscle group stimulated by an electrical stimulus. To derive the model, the dynamic equilibrium of the moment acting on the knee joint was first described. The damping and stiffness properties of the knee were calculated as well as the gravitational and inertial characteristics of the anatomical segments. To estimate the unknown viscous-elastic parameters, a passive pendulum test of the lower limb was carried out. In addition, the knee movements which resulted from stimulating the quadriceps muscle group with electrical stimulus were recorded. The autoregressive with exogenous inputs (ARX) approach was used to estimate a transfer function that describes the relationship between the electrical stimuli on the leg as an input and the resultant knee joint extension torque as an output, as

$$H(s) = \frac{G}{1 + s\tau} \quad (1)$$

where τ is the time constant and G is the static gain. In this work, an average value of knee joint's viscous coefficient for paraplegic subjects (0.287 N m s/rad) is added to the knee joint of the humanoid model. Also the static gain (0.04 N m/ μ s) and time constant (0.45 s) values were chosen as provided in Ref. [40].

3. Control strategy

The torque generated by a muscle, as a response to an FES signal, can be controlled by varying the pulse width of the stimulus in a closed-loop control approach. Knee angle reference for 35 rpm speed, the lowest cycling speed used clinically [19], is used in this work. The actual knee trajectory, using a position sensor in the humanoid–bicycle model, is measured. The knee trajectory is used as feedback and compared with the knee angle reference signal. According to the error signal, the controllers adjust the pulse width of the stimulus to control muscle torque and consequently control leg pedalling movement to follow the reference.

3.1. FES-cycling by stimulating the quadriceps without assist mechanism (scenario I)

The aim here is to perform FES driven cycling for paraplegics by stimulating single muscle group, the quadriceps, of each leg without using any assist mechanism. By stimulating the quadriceps, only knee extension, i.e. pushing, torque can be generated. Knee extension torque is essential to speed up the pedalling movement. In order to maintain a required cadence, i.e. pedalling speed, it is important to provide an opposite torque to resist and retard the movement in case the leg speed exceeds the desired cadence. Since no flexion muscle is stimulated in this work, the only way to provide an opposite torque is by stimulating the quadriceps of the parallel leg within a specific period at which the knee extension of the parallel leg leads to retarding rather than speeding up the movement. For this reason, each leg's movement is divided into three phases; pushing, resisting and resting as shown in Fig. 3.

The pushing phase is the period at which the quadriceps of the corresponding leg is stimulated to generate torque to speed up the

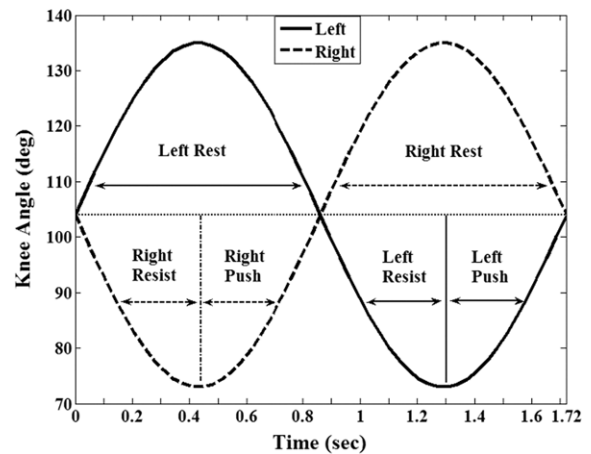


Fig. 3. Cycling phases described according to the knee trajectory of right and left leg.

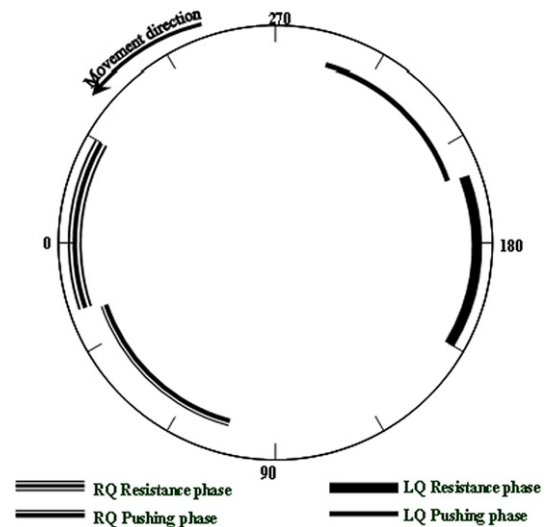


Fig. 4. Cycling phases defined according to crank angle (RQ = Right Quadriceps, LQ = Left Quadriceps) [30]. The corresponding knee angles: ResistPhase = 80°–73° one leg, 127°–134° parallel leg (Rest Phase); PushPhase = 73°–96° one leg, 134°–111° parallel leg (Rest Phase).

movement. The resist phase is the period at which the quadriceps is stimulated to retard the movement. While the rest phase is the period at which the muscle is not stimulated and left to rest before the next stimulus is due. Hence, the quadriceps muscle group of each leg is stimulated twice per cycle, once to speed up the leg, and then to retard or resist the movement. Those three phases are interpreted and defined according to crank angle as shown in Fig. 4.

In this work, as the cycling ergometer is at the same horizontal level of the hip joint, as in Fig. 5, the cycling dead spots, at which the leg movement changes direction and causes significant increase in the angular velocity, appear around (0° and 180°) of the crank angle, i.e. 72° and 135° of right and left knee angles. For this reason the resisting phase is defined around 0° and 180° to reduce the effect of the dead spots and prevent jerking during the cycling.

3.1.1. Closed-loop control using fuzzy logic

Fuzzy logic control (FLC) is well known for its effectiveness in controlling nonlinear systems and is a model-free mechanism based on linguistics rather than on mathematics. In this work fuzzy logic is implemented to control and achieve FES-cycling by stimulating the quadriceps muscle only. The closed-loop control structure used is shown in Fig. 6. The controllers are utilized to change

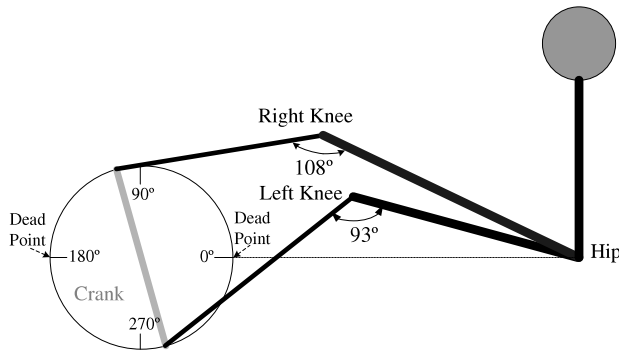


Fig. 5. The dead points and the crank position with respect to hip joint.

the pulse width of the stimulus applied to the muscle to adjust the amount of the generated muscle force required to maintain a desired speed. The crank angle is used to specify the phases for each leg, previously explained in Section 3.1.

Since the amounts of push and resist, required to maintain a desired speed, are not equal, two different controllers are used for each leg. Right and left knee angle reference of 35 rpm cycling speed is used to compare with the actual knee trajectory signal taken from a position sensor located in the humanoid–bicycle model. The difference, i.e. error, between these signals and the derivative, i.e. rate of change of error, are used by the fuzzy controller of each leg to accordingly adjust the pulse width of the stimulus.

Each controller has three inputs and one output. The first two inputs, the error and the rate of change of error, are normalized by input scaling factors (G_1 , G_2 , G_4 , G_5 , etc.), while the third input is the crank angle which is measured by a position sensor in the humanoid–bicycle model. The two normalized FLC inputs (error and rate of change of error) are fuzzified using fuzzy set of five equally distributed, with 50% overlapping, Gaussian membership functions. While the third FLC input, the crank angle which is used to achieve the synchronization among the four controllers by dividing it into phases, is fuzzified using fuzzy set of four variables (RR, RP, LR and LP) that are defined using trapezoidal membership functions to ensure minimum overlapping among the defined phases. The fuzzy output, which results from the fired fuzzy rules of the FLC, changes to crisp values using the centre of area defuzzification method. The output is defuzzified using five equally distributed

Gaussian membership functions. The fuzzy input/output membership functions of each controller are depicted in Figs. 7 and 8.

The standard twenty five fuzzy rules are used for each controller, as shown in Table 4, in addition to the third input which allows the activation of each controller during its own phase. Since the quadriceps is the only muscle group stimulated in this work, which can produce extension torque only, i.e. cannot produce negative or flexion torque, the negative action of the controller is prevented using a saturation block.

The input and output scaling factors of each controller are obtained heuristically. The values of the scaling factors used are: $G_1 = G_{10} = 0.008$, $G_2 = G_{11} = 0.0032$, $G_4 = G_7 = 0.016$, $G_5 = G_8 = 0.0034$, $G_3 = G_6 = G_9 = G_{12} = 900$.

3.1.1.1. Results. It is obvious from Figs. 9 and 10 that the proposed control strategy was successful in achieving coordinated leg cycling movement and acceptable tracking performance. At the beginning of the cycling the tracking error was extremely high; see Fig. 11. The reason behind this is that the starting position of the leg was chosen to be at 110° of the crank angle, at which the leg muscle is not stimulated due to being at the resting phase as previously shown in Fig. 7(c), in order to benefit from the gravitational force on the leg to start the movement and overcome the inertia. Despite the initial large error, the controllers were successful in minimizing the error in successive cycles. However, the cycling cadence was not steady at 210 deg/s (i.e. 35 rpm), as shown in Fig. 12. This is due to the effect of the cycling dead points that caused the rapid change in the angular velocity of the crank.

The muscle torque of each leg and the pulse width regulated by the FLC unit of each phase can be seen in Figs. 13–18. It is clear that the torques produced by both legs are not equal, which implies that one of the legs has received more stimulation than the other. This is due to the fact that the input/output scaling factors of the controllers, as well as the firing angles of the defined phases, were chosen heuristically and will need fine tuning for both legs to receive equal amount of exercise and to equally contribute in the cyclic pedalling motion. Moreover, it is clear that the quadriceps muscle of each leg is stimulated twice per cycle, one to speed up, i.e. push, and the other to retard, i.e. resist, the movement. Due to the tracking delay at the beginning of the cycling, resulting from the free-fall of the right leg, the stimulation intensity during the pushing phase of the left leg was relatively high, Figs. 15 and 16. As the resist action took place in successive cycles, Figs. 17 and 18, the tracking error was reduced and hence the stimulation intensity.

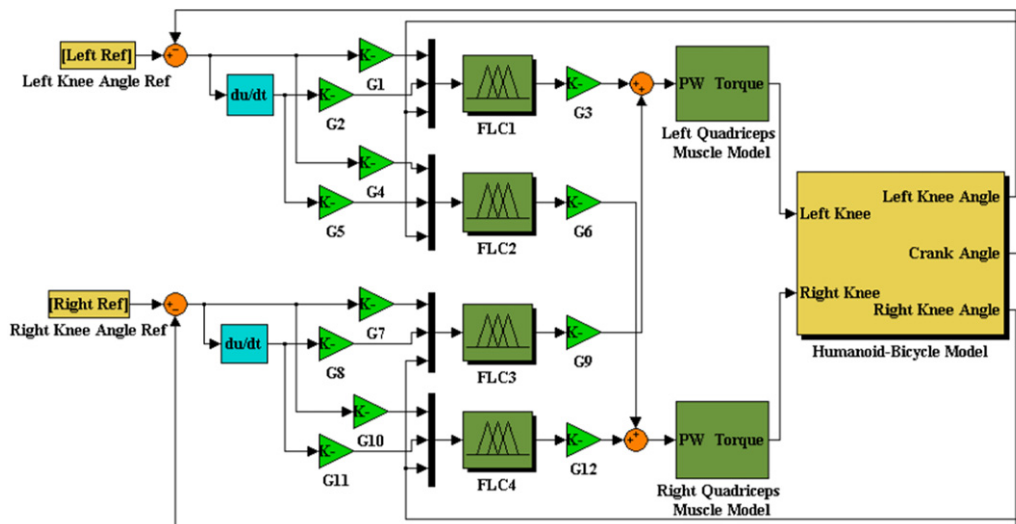


Fig. 6. The control structure using fuzzy logic controllers.

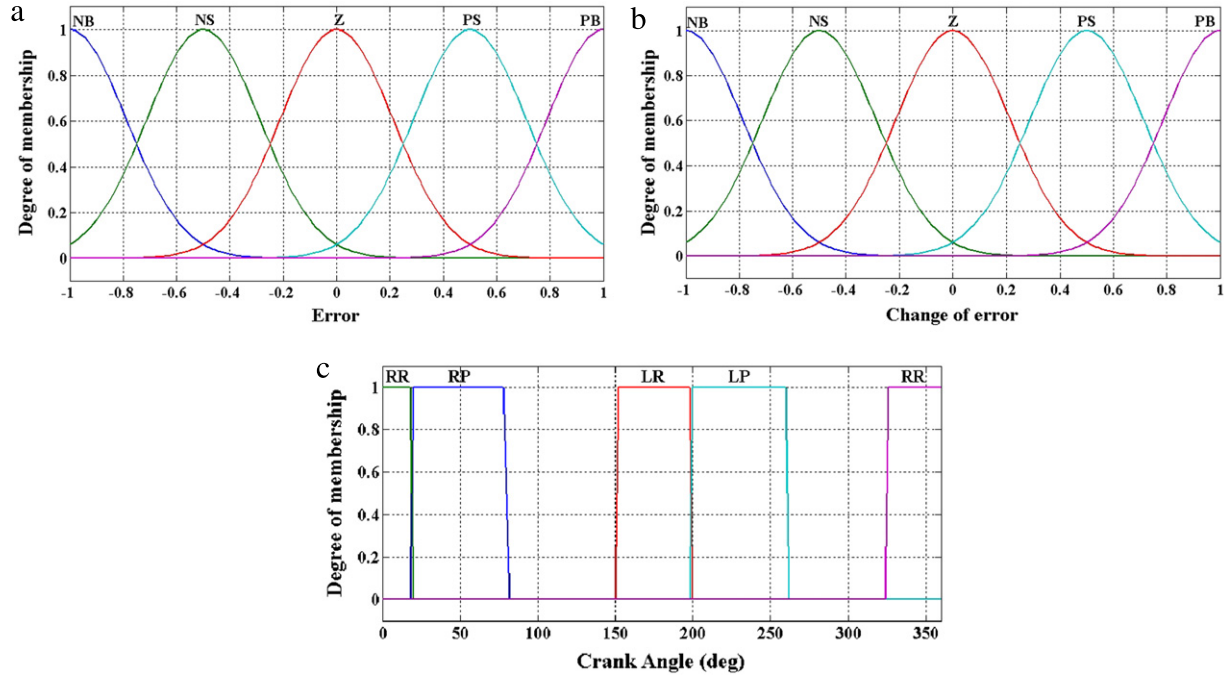


Fig. 7. Controller's input membership functions; the third input (c) shows the phases (RR = right resist, RP = right push, LR = left resist, LP = left push).

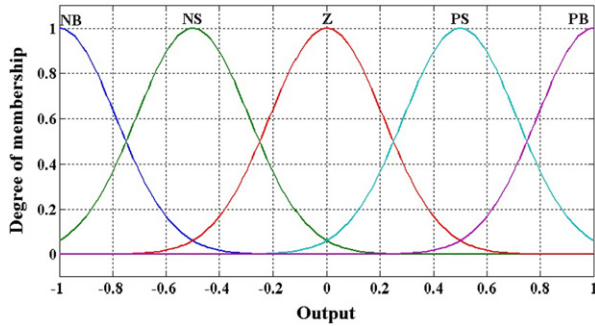


Fig. 8. Controller's output membership functions.

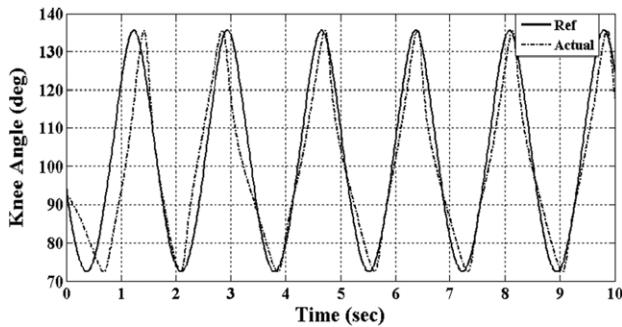


Fig. 9. Left leg tracking the reference.

It is worth mentioning that successive muscle stimulation leads to fatigue the muscle rapidly as the muscle re-contracts and consumes more energy. Hence, in this scenario, as each muscle is stimulated twice per cycle it is expected to fatigue the muscle rapidly and consequently terminate the training session after a short period.

3.2. FES driven cycling using the flywheel and electrical clutch mechanism (scenario II)

To reduce the probability of rapid muscle fatigue that might result from successive stimulation to the muscle (as in the previous

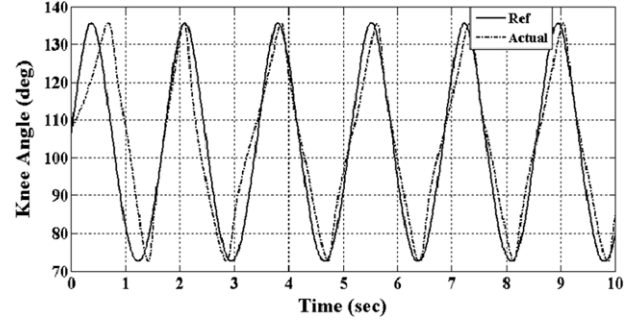


Fig. 10. Right leg tracking the reference.

scenario), an energy storage device, i.e. a flywheel, with an electrically activated clutch mechanism is utilized. The mechanism is used to assist and retard the cycling when necessary and to replace muscle stimulation in the resist phase. The electrical clutch engages the flywheel with the crank to absorb the excessive energy, produced by the leg, to store it as kinetic energy in the flywheel. Also the flywheel, loaded with energy, is engaged by the clutch to support the cycling in case the energy of the leg is not enough to pedal.

The flywheel stores the rotational kinetic energy according to the following equation:

$$E_k = \frac{1}{2} I \omega^2 \quad (2)$$

$$I = \frac{1}{2} m r^2 \quad (3)$$

where

E_k : rotational kinetic energy [J],

I : rotational inertia of the flywheel about the rotating axis [kg m²],

ω : angular velocity of the flywheel [rad/s],

m : mass of the flywheel [kg],

r : radius of the flywheel [m].

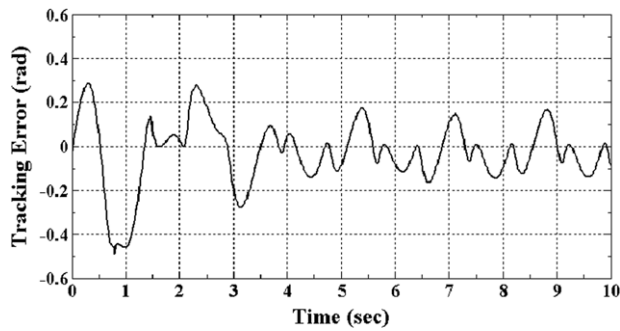


Fig. 11. Right leg tracking error.

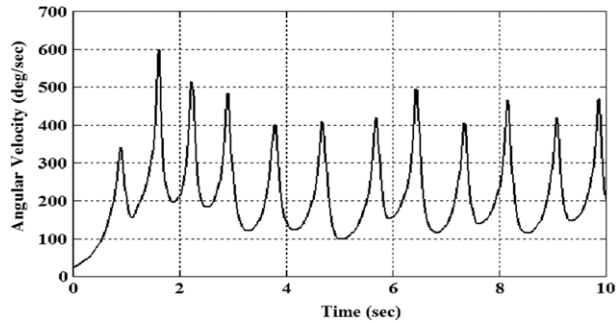


Fig. 12. Crank angular velocity.

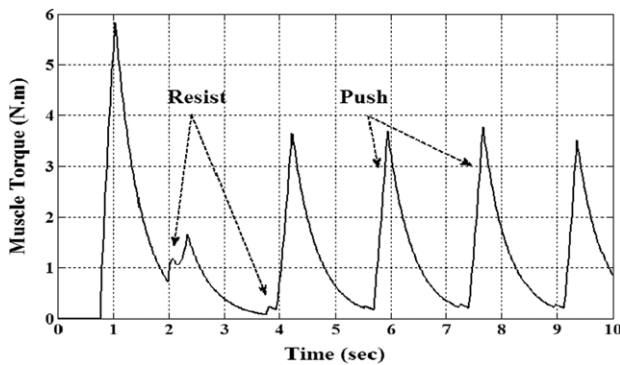


Fig. 13. Muscle torque of left leg.

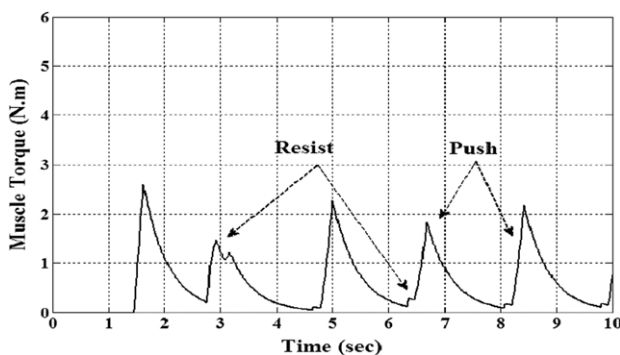


Fig. 14. Muscle torque of right leg.

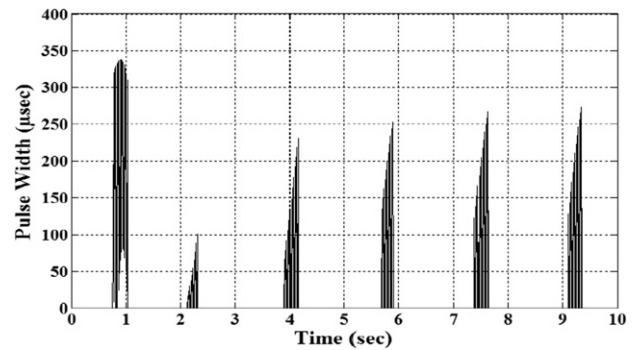


Fig. 15. Pulse width in Left-Push phase.

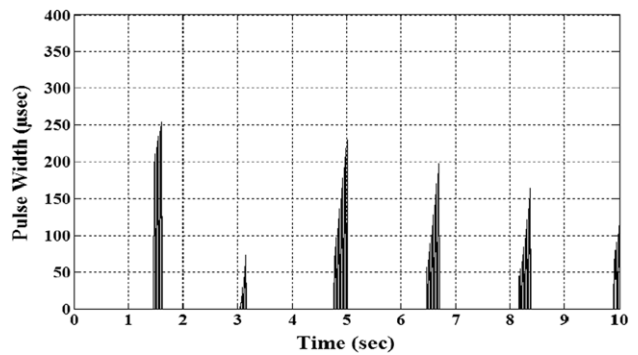


Fig. 16. Pulse width in Right-Push phase.

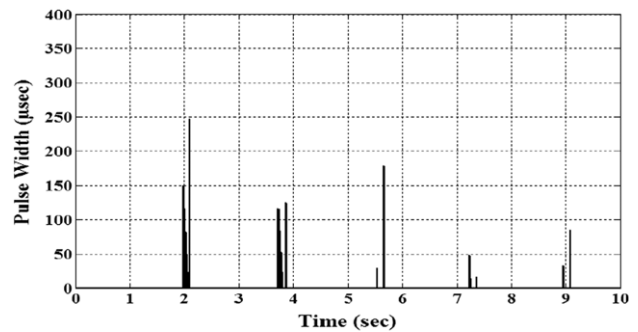


Fig. 17. Pulse width in Left-Resist phase.

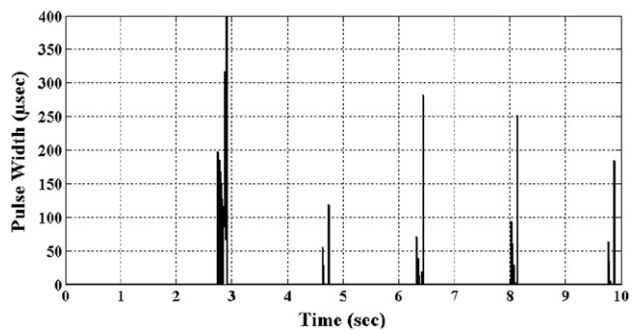


Fig. 18. Pulse width in Right-Resist phase.

The above equations imply that the maximum amount of kinetic energy that the flywheel can store depends on the mass, the size and the angular velocity of the flywheel.

In this scenario, only two FLCs are used to maintain speeding up the cycling during the push phase, while the resist phase is replaced by the flywheel mechanism. As in the previous scenario, each controller has three inputs; the error, the rate of change of

error and the crank angle. The difference between the knee reference and the actual knee trajectory, i.e. error, and its derivative, i.e. rate of change of error, are used by the controllers to adjust the pulse width of the stimulus, while the crank angle is used to specify the period at which the controllers are active. The input and output scaling factors used in this scenario are the same as those used in scenario I ($G1 = G10 = 0.008$, $G2 = G11 = 0.0032$, $G3 = G12 = 900$). The control block diagram is shown in Fig. 19.

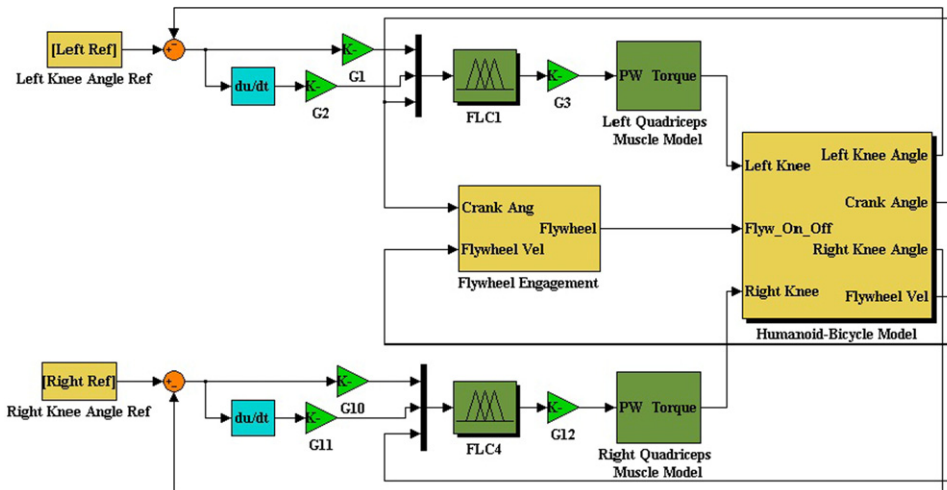


Fig. 19. Control block diagram using flywheel and electrical clutch mechanism.

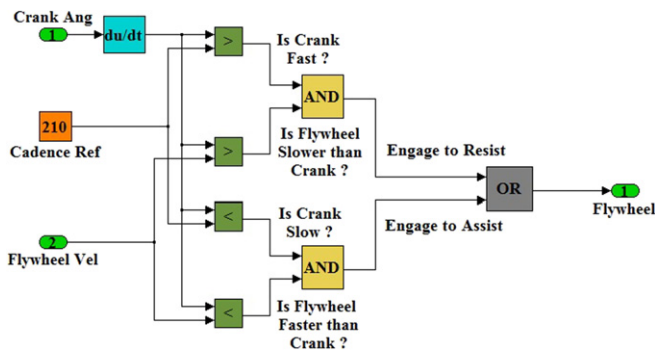


Fig. 20. The logic of the flywheel engagement mechanism.

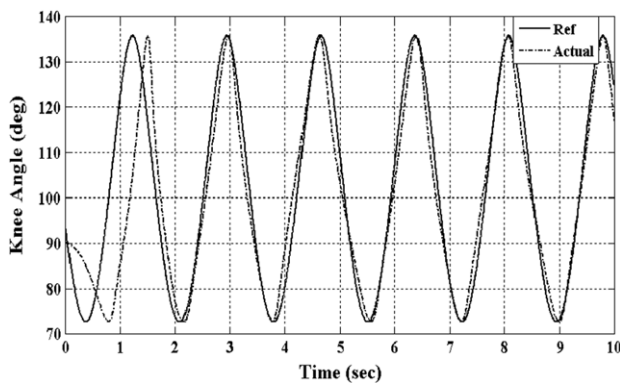


Fig. 21. Left leg tracking the reference.

The engagement and disengagement of the flywheel is achieved through an electrical clutch of an on/off control input. The flywheel's engagement and disengagement process was controlled according to two factors; the first is whether the crank's angular velocity is higher or lower than the desired cadence, and the second is whether the flywheel's angular velocity is less or greater than that of the crank. If the crank's speed is higher than the required speed, i.e. the tracking error is negative, and the flywheel's speed is less than the speed of the crank, i.e. the flywheel has the ability to resist the movement, the clutch will engage the flywheel with the crank to absorb the surplus in the energy and store it as kinetic energy, and produce a damping effect on the movement. If the crank's speed is less than the required speed, i.e. the tracking error is positive, and the flywheel's speed is higher than that of the crank, i.e. the flywheel has the ability to assist the leg, the flywheel will

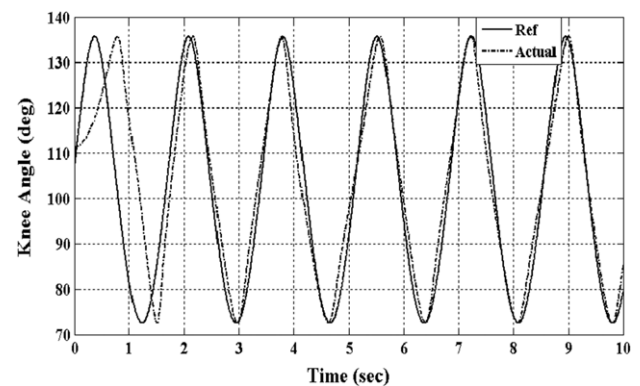


Fig. 22. Right leg tracking the reference.

be engaged to assist and speed up the cycling by discharging its kinetic energy into the system. The logic implemented for flywheel's engagement and disengagement via an electrical clutch is shown in Fig. 20. The angular velocity of the flywheel is measured by a velocity meter in the humanoid-bicycle model provided by Vn4D software, while the derivative of the crank angle, provided by the same software, is used to measure the angular velocity of the crank.

To illustrate the use of this mechanism, a simulation video is available and accompanies the electronic version of this manuscript. Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.robot.2013.10.005>.

3.2.1. Results

The tracking performance of right and left leg, tracking error of right leg and the flywheel's engagement and disengagement periods can be seen in Figs. 21–25. Although the control strategy produced acceptable tracking performance and coordinated pedalling movement, it is obvious from Figs. 21 and 22 that there was a slight delay in the tracking at the first cycle. This is due to the fact that the flywheel assist mechanism was activated after two seconds from the start to benefit from the rotational momentum caused by the gravitational force on the leg. Even though, the mechanism with the controller was successful in following the reference in subsequent cycles.

From Figs. 26 and 27 it is obvious that the flywheel retarded the movement by absorbing the crank's energy and then speeded up the movement by releasing the stored kinetic energy into the crank. Also, as a comparison with scenario I (Figs. 12 and 26), it can be noticed that the flywheel reduced the fluctuation in the crank angular velocity, by suppressing the rapid changes in the angular

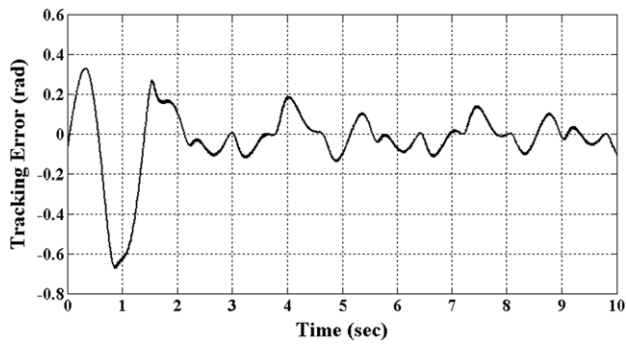


Fig. 23. Right leg tracking error.

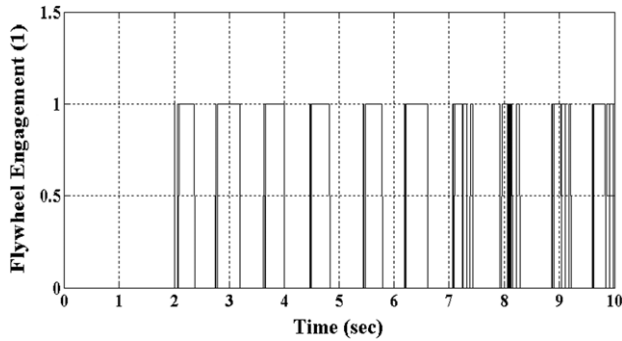


Fig. 24. Flywheel engagement (Resist periods).

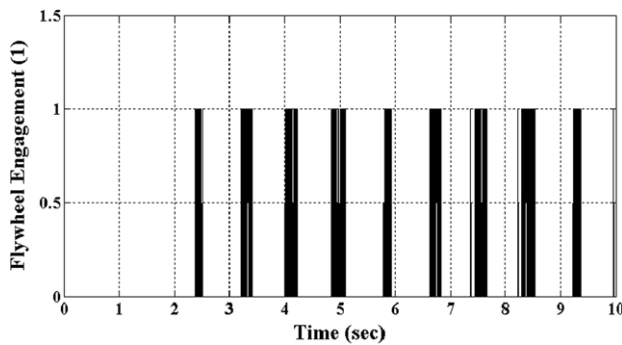


Fig. 25. Flywheel engagement (Assist periods).

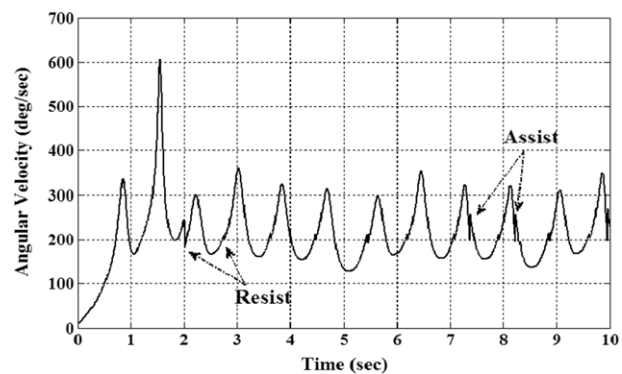


Fig. 26. Crank angular velocity.

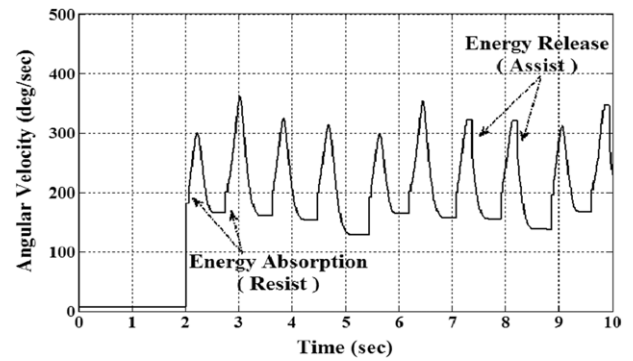


Fig. 27. Flywheel angular velocity.

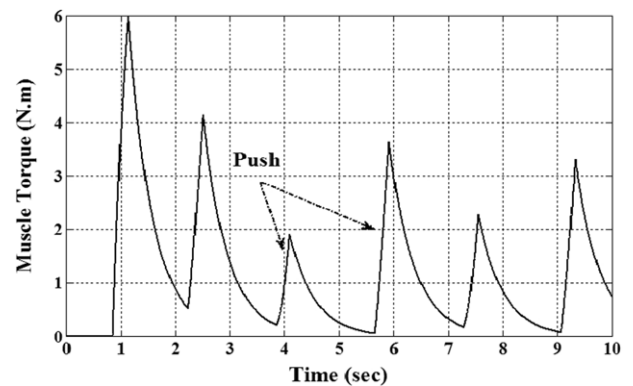


Fig. 28. Left muscle torque.

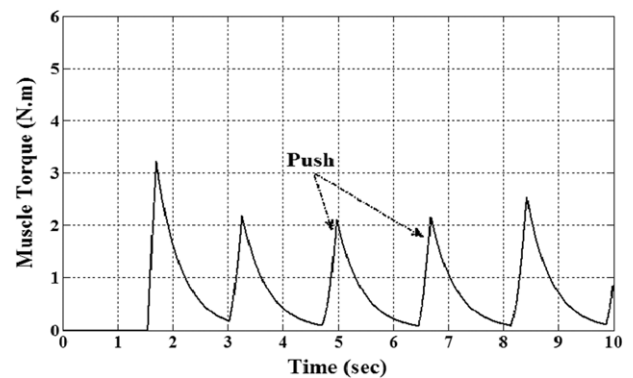


Fig. 29. Right muscle torque.

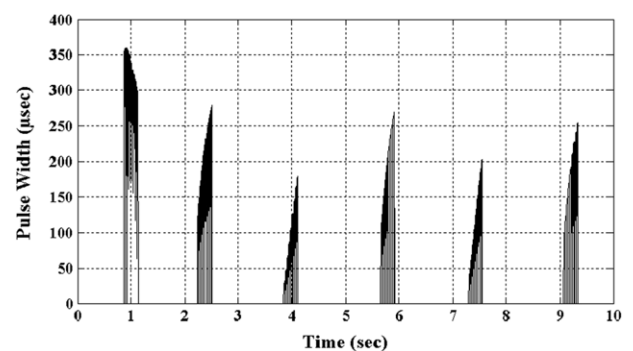


Fig. 30. Pulse width of left muscle.

velocity at the dead points, and produced smoother and closer to the desired, i.e. 210 deg/s, cadence.

It is clear from Figs. 28 and 29 that the muscles were stimulated only once per cycle during the pushing phase and there were no successive stimulations as appeared in scenario I. However, from Figs. 30 and 31 it can be seen that the stimulation intensity slightly increased as compared with that of scenario I. The reason for

this increase in the stimulation intensity is that the flywheel has slightly slowed down the speed (Figs. 12 and 26) in the system and imposed a slight load on the crank. Thus, the controllers, in turn, slightly increased the stimulation intensity on the legs to

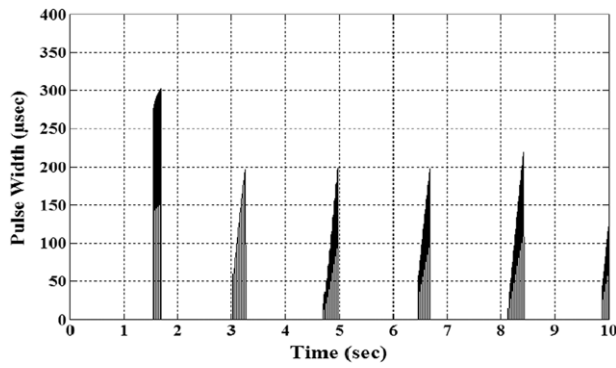


Fig. 31. Pulse width of right muscle.

speed up the movement. However, since the flywheel replaces the stimulation in the resist phase, as appeared in scenario I, in this case the muscle is stimulated only once per cycle, during the pushing phase, and hence this reduces the muscle stimulation period, in comparison with that of scenario I, and consequently allows more time for the muscle to rest before next stimulus is due.

From fatigue point of view, as it is difficult to compare between the increased/decreased pulse width in both scenarios (Figs. 15–18, 30 and 31), the need for a force-drop indicator was raised to assess the two scenarios.

4. Force-drop indicator

As can be seen from the results of scenario I, each leg muscle was stimulated twice per cycle, while in scenario II the stimulation dropped to one stimulus per cycle but at the same time the stimulation intensity in scenario II slightly increased as compared with that of scenario I. For these reasons, it is difficult to compare the two scenarios and anticipate the improvement from fatigue point of view without using an indicator that takes into consideration the amount of stimulus as well as the pulse width of the signal applied to the muscle. Since the muscle model utilized in this work has no fatigue index, a need to develop a simple force-drop indicator, to be used for comparison purposes, arose.

Clinically recorded data was used to develop the force-drop indicator. The experiment was carried out with the aid of a paraplegic subject of incomplete spinal cord lesion T2–T3. The subject was seated in semi-upright position (45°–60° hip angle) with the thigh fixed to the seat (80°–90° knee angle), with thigh supporters, to allow free leg movement. In addition to Velcro straps were used to support and stabilize the trunk and waist during the experiment to prevent any external influence on the quadriceps response to the FES signal.

Muscle stimulation was performed using a RehaStim Pro 8 channels (Hasomed GmbH, Germany) stimulator, which received its commands from Matlab software through USB connection. Electrical stimulus was delivered to the quadriceps through two Multistick gel surface electrodes (Pals platinum, Axelgaard, USA, size: 50 mm × 90 mm). The cathode (–) was placed over the motor point of the rectus femoris (proximal to trunk) while the anode (+) was placed above the patella (distal to trunk). To find out the optimal location of the cathode, i.e. to obtain the highest muscle contraction, the electrode was moved around the skin over the motor point using the same stimulation signal for all trials, with the knee almost fully extended. To record the muscle force resulting from FES signals, a force transducer (PCE-FM200, PCE group company, Deutschland) was used. The force transducer was placed about 4 cm proximal to the lateral malleolus, against the anterior aspect of the leg [44] through a padded cuff equipped with a hook, as shown in Fig. 32. The force transducer was fixed in housing for

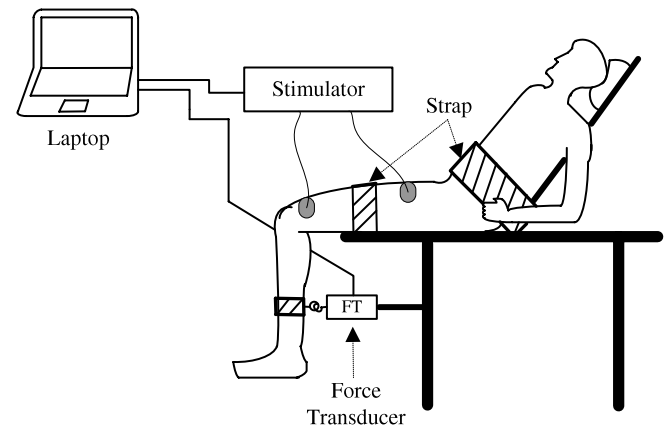


Fig. 32. Experimental set-up to record quadriceps isometric contraction force in response to the FES signal.

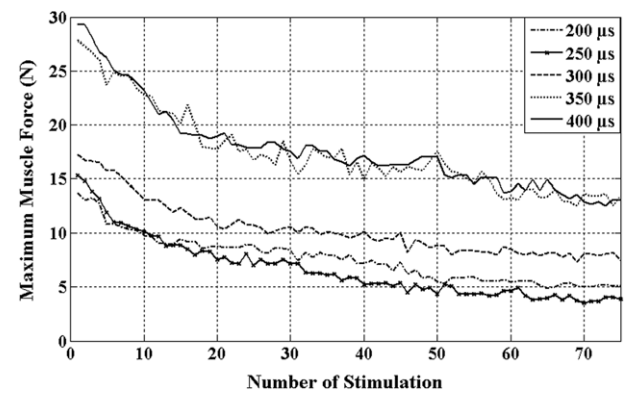


Fig. 33. Peak muscle force for 75 stimulations of different pulse widths.

measuring the isometric contraction forces resulting from stimulating the quadriceps. The force transducer was connected to a computer through an RS-232 port to simultaneously record muscle force resulting from FES signals. The isometric fatigue test was previously used to test muscle performance, before and after load, in non-isometric, i.e. isotonic and isokinetic, cycling exercise [45,46].

The test was performed by applying one stimulus per ten seconds (3 s on and 7 s off), with different pulse widths (200–400 µs) while keeping other parameters fixed (current 40 mA, frequency 30 Hz). The maximum muscle force recorded during the isometric test for different pulse widths can be seen in Fig. 33.

To derive the force-drop monitor, first of all, each line of the data obtained, Fig. 33, is normalized, i.e. subtracting each resultant force point from the maximum force obtained from that pulse width and then dividing the result with the same maximum force. This operation calculates the rate of force drop in the muscle stimulated with a specific pulse width. The resultant normalized force-drop data together with the number of stimulus and the pulse width are utilized, with the aid of curve fitting toolbox, to derive the indicator as shown in Fig. 34.

Linear 3rd order polynomial fitting is implemented with 9 coefficients to increase the accuracy of the model; the resultant relationship combines the pulse width (Y) and the number of stimulus (X) with the resulted muscle force (F) as

$$F(x, y) = P_{00} + P_{10} * X + P_{01} * Y + P_{20} * X^2 + P_{11} * X * Y + P_{02} * Y^2 + P_{30} * X^3 + P_{21} * X^2 * Y + P_{12} * X * Y^2. \quad (4)$$

The coefficients (with 95% confidence bounds) of Eq. (4) are shown in Table 5. The goodness of the fit is assessed as acceptable according to the statistics shown in Table 6.

Eq. (4) is used to assess the performance of both approaches in scenario I and scenario II and assess the benefits of the proposed

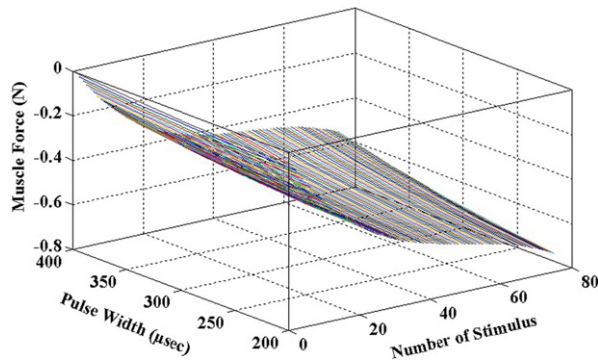


Fig. 34. The derived force-drop monitor.

Table 4

Fuzzy rules.

e	Δe				
	NB	NS	Z	PS	PB
NB	NB	NB	NB	NS	Z
NS	NB	NB	NS	Z	PS
Z	NB	NS	Z	PS	PB
PS	NS	Z	PS	PB	PB
PZ	Z	PS	PB	PB	PB

Table 5

Force-drop indicator's coefficients.

Coefficient	Value
P_{00}	0.01435
P_{10}	-0.02739
P_{01}	-0.0004771
P_{20}	0.0004631
P_{11}	2.976e-006
P_{02}	1.176e-006
P_{30}	-3.052e-006
P_{21}	-2.456e-008
P_{12}	1.291e-008

Table 6

Statistics for goodness-of-fit assessment.

Statistic	Value
Sum of Squares Due to Error (SSE)	1.552
R-Squared Error	0.8531
Adjusted R-Squared	0.8499
Root Mean Square Error (RMSE)	0.06512

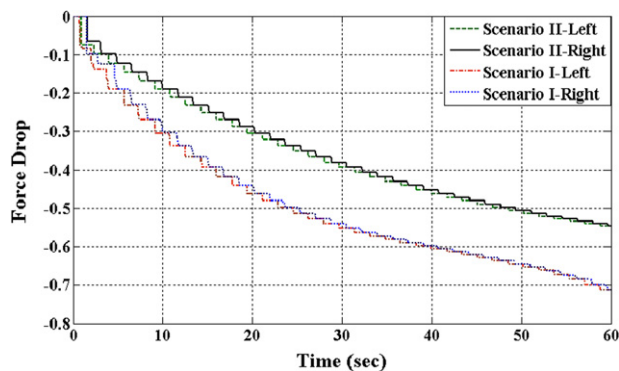


Fig. 35. Force drop in right and left leg muscles in both scenarios.

cycling assist mechanism from fatigue point of view. After applying the derived indicator in both scenarios, as in Fig. 35, it is obvious that the force drop in scenario II was slower and delayed by approximately 14%–17% as compared with scenario I, as in Fig. 36.

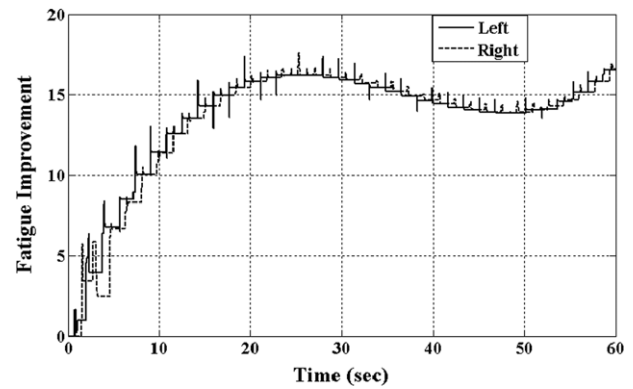


Fig. 36. Fatigue improvement percentage of scenario II as compared with scenario I.

5. Discussion and conclusion

A novel assist mechanism for FES-assisted cycling has been presented. The introduced mechanism, represented by a flywheel and an electrical clutch, has been utilized in FES-cycling application for the first time. Individuals with spinal cord injury who have weak muscles may encounter difficulties to overcome the high inertia of the ergometer with fixed-engaged flywheel; therefore, assistance from the therapist, or an external device like a motor, is required to initiate the movement. With the use of an electrical clutch, the flywheel will not be permanently engaged with the system and hence will not impose extra load on the muscle especially at the initial stage of the cycling. The flywheel, as an energy storage device, together with an electrical clutch can be used to absorb the excess energy in the system, store it as kinetic energy and reuse the same energy to assist the leg.

This mechanism has been utilized to assist the leg in FES-cycling by stimulating single muscle group, the quadriceps. To perform smooth and coordinated FES-cycling by stimulating the quadriceps only, the muscle should be stimulated twice per cycle, push phase and resist phase, to govern a specific cadence. However, with the new proposed mechanism the muscle is stimulated once per cycle and the cycling dead spots, 0° and 180° of the crank, were passed smoothly due to the use of the flywheel.

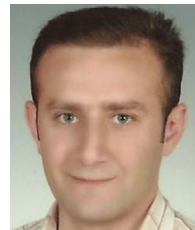
It has been noticed from the results that the stimulation intensity has slightly increased with the new mechanism, even though the derived force-drop indicator shows that the new mechanism delays the fatigue by approximately 14%–17% as compared with FES driven cycling without assistance. As a result, it can be concluded that the new mechanism is promoting prolonged FES-cycling session and extended work rate for both legs.

Despite the promising results obtained, it is worth mentioning that the controllers' input/output scaling factors have been obtained heuristically. Besides the scaling factors, several parameters, such as flywheel's weight and size, gear ratio between the flywheel and the crank, and the position of the crank in relation to the hip joint, play an essential role and need to be optimized, for optimal tracking performance and assistance, before the hardware implementation of the system.

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